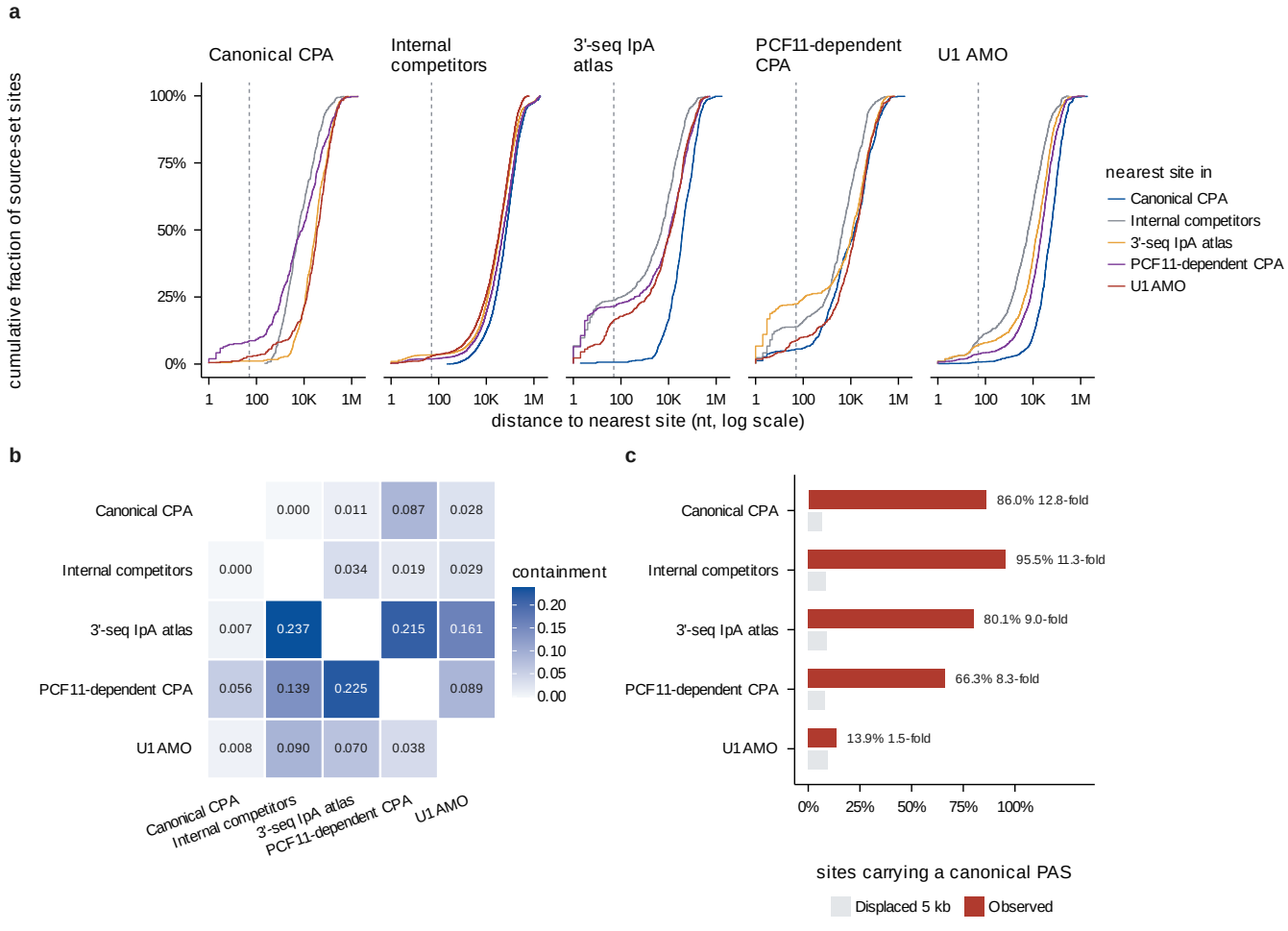

a
Canonical CPA
Internal competitors
3'-seq IpA atlas
PCF11-dependent CPA
U1 AMO
cumulative fraction of source-set sites
100%
75%
50%
25%
0%
1
100
10K
1M
distance to nearest site (nt, log scale)
nearest site in
Canonical CPA
Internal competitors
3'-seq IpA atlas
PCF11-dependent CPA
U1 AMO
b
Canonical CPA 0.000 0.011 0.087 0.028
Internal competitors 0.000 0.034 0.019 0.029
3'-seq IpA atlas 0.007 0.237 0.215 0.161
PCF11-dependent CPA 0.056 0.139 0.225 0.089
U1 AMO 0.008 0.090 0.070 0.038
containment
0.20
0.15
0.10
0.05
0.00
c
Canonical CPA 86.0% 12.8-fold
Internal competitors 95.5% 11.3-fold
3'-seq IpA atlas 80.1% 9.0-fold
PCF11-dependent CPA 66.3% 8.3-fold
U1 AMO 13.9% 1.5-fold
0%
25%
50%
75%
100%
sites carrying a canonical PAS
Displaced 5 kb
Observed
d
Random forest, probability
CNN, percentile of all windows
Canonical CPA 0.95 0.88
Internal competitors 0.91 0.49
3'-seq IpA atlas 0.84 0.64
PCF11-dependent CPA 0.81 0.75
U1 AMO 0.71 0.63
0.00
0.25
0.50
0.75
1.00
score